Analysis and Visualization

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analytical evaluation

Error Analysis



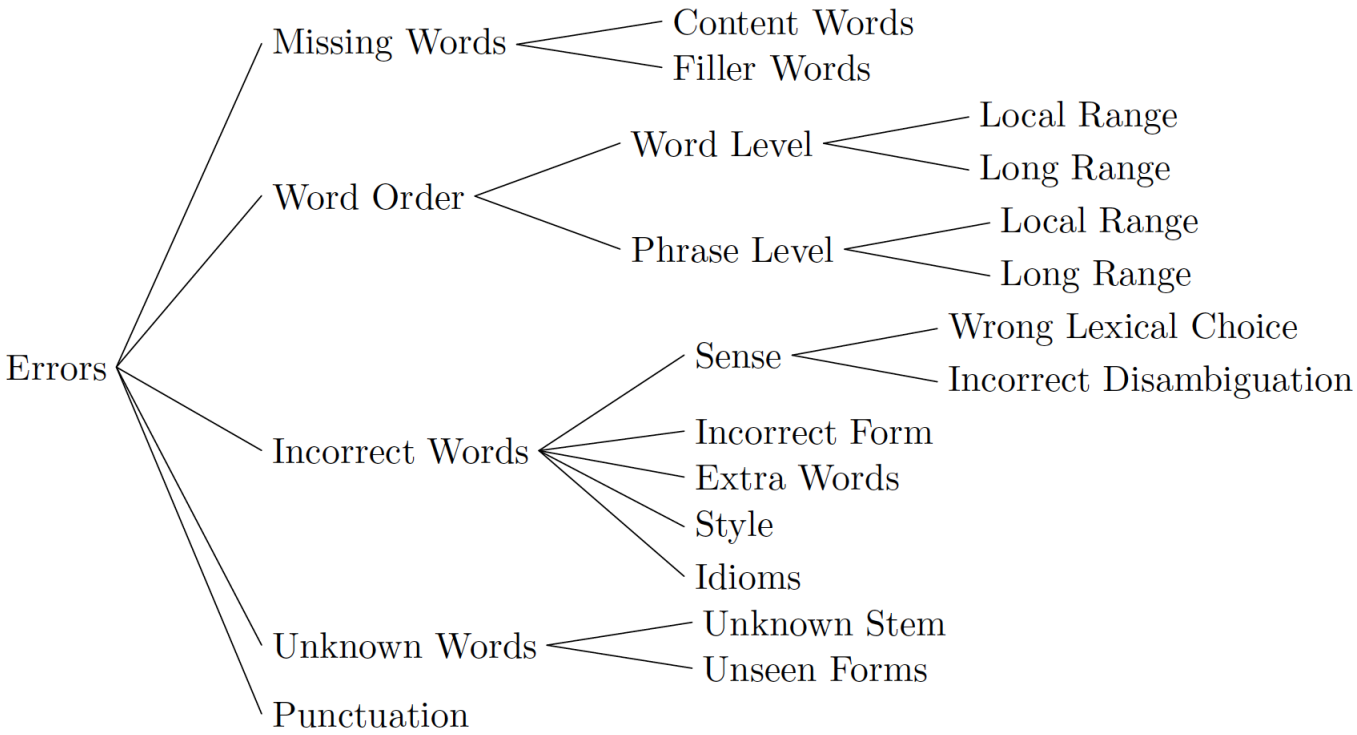
- Manually inspect output of machine translation system
- Identify errors and categorize them
- Specific problems of neural machine translation
  - dropped input / added output
  - gibberish (*the the the the*)
  - hallucinated output

Hallucinated Output

- Examples of extreme translation failures
  - Low resource example
    Republican strategy to counter the re-election of Obama
    Un órgano de coordinación para el anuncio de libre determinación
  - Out of domain example
    Schaue um dich herum.
    EMEA / MB / 049 / 01-EN-Final Work progamme for 2002

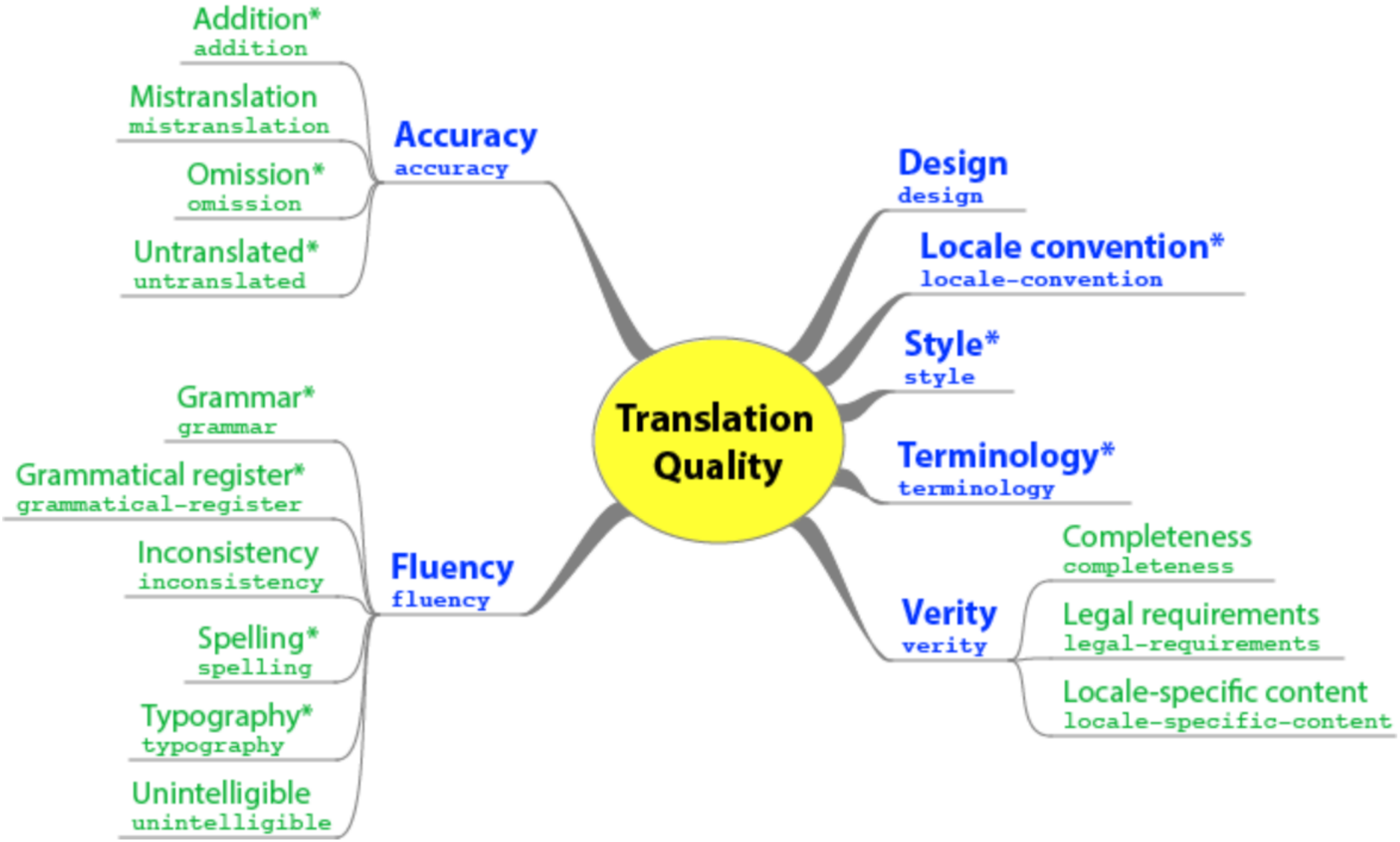
- Neural MT goes off track
  - turns into generative language model
  - ignores input context

Linguistic Categories



"Error Analysis of Statistical Machine Translation Output" (Vilar et al., LREC 2006)

MQM


Translation Quality
Accuracy
accuracy
Addition*
addition
Mistranslation
mistranslation
Omission*
omission
Untranslated*
untranslated
Fluency
fluency
Grammar*
grammar
Grammatical register*
grammatical-register
Inconsistency
inconsistency
Spelling*
spelling
Typography*
typography
Unintelligible
unintelligible
Design
design
Locale convention*
locale-convention
Style*
style
Terminology*
terminology
Verity
verity
Completeness
completeness
Legal requirements
legal-requirements
Locale-specific content
locale-specific-content

Bentivogli et al. (EMNLP 2016)



- Manually corrected machine translation
- Breakdown of word edits
  - by part-of-speech tag
  - multi-word construction, e.g., AUX:V constructions such as *can eat*
- Systems
  - NMT: neural machine translation
  - PBSY: phrase-based statistical
  - HPB: hierarchical phrase-based statistical
  - SPB: syntax-based statistical

Class	NMT-vs-PBSY	NMT	PBSY	HPB	SPB
V	-70%	35	116	133	155
PRO	-57%	22	51	53	62
PTKZU	-54%	6	13	4	11
ADV	-50%	14	28	44	36
N	-47%	37	70	99	56
KON	-33%	6	9	8	12
PREP	-18%	18	22	27	28
PTKNEG	-17%	10	12	10	7
ART	-4%	26	27	38	35
aux:V	-87%	3	23	17	18
neb:V	-83%	2	12	7	19
objc:V	-79%	3	14	21	24
subj:PRO	-70%	12	40	34	46
root:V	-68%	6	19	28	27
adv:ADV	-67%	8	24	33	28
obja:N	-65%	6	17	28	12
cj:V	-59%	7	17	21	22
part:PTKZU	-54%	6	13	4	11
obja:PRO	-38%	5	8	14	7
mroot:V	-36%	7	11	26	20
pn:N	-36%	16	25	33	19
subj:N	-33%	6	9	10	7
pp:PREP	-30%	14	20	19	23
adv:PTKNEG	-17%	10	12	10	7
det:ART	-4%	26	27	38	34
all	-48%	222	429	493	488

targeted test sets

Challenge Set

- Create challenging sentence pairs with specific problems

Src	The repeated calls from his mother **should** have alerted us.
Ref	Les appels répétés de sa mère **auraient** dû nous alerter.
Sys	Les appels répétés de sa mère devraient nous avoir alertés.
Is the subject-verb agreement correct (y/n)? **Yes**	

- "A Challenge Set Approach to Evaluating Machine Translation" (Isabelle et al., EMNLP 2017)

Challenge Set: Results

Category	Subcategory	#	PBMT-1	NMT	Google NMT
Morpho-syntactic	Agreement across distractors	3	0%	100%	100%
	through control verbs	4	25%	25%	25%
	with coordinated target	3	0%	100%	100%
	with coordinated source	12	17%	92%	75%
	of past participles	4	25%	75%	75%
	Subjunctive mood	3	33%	33%	67%
Lexico-syntactic	Argument switch	3	0%	0%	0%
	Double-object verbs	3	33%	67%	100%
	Fail-to	3	67%	100%	67%
	Manner-of-movement verbs	4	0%	0%	0%
	Overlapping subcat frames	5	60%	100%	100%
	NP-to-VP	3	33%	67%	67%
	Factitives	3	0%	33%	67%
	Noun compounds	9	67%	67%	78%
	Common idioms	6	50%	0%	33%
	Syntactically flexible idioms	2	0%	0%	0%
Syntactic	Yes-no question syntax	3	33%	100%	100%
	Tag questions	3	0%	0%	100%
	Stranded preps	6	0%	0%	100%
	Adv-triggered inversion	3	0%	0%	33%
	Middle voice	3	0%	0%	0%
	Fronted should	3	67%	33%	33%
	Clitic pronouns	5	40%	80%	60%
	Ordinal placement	3	100%	100%	100%
	Inalienable possession	6	50%	17%	83%
	Zero REL PRO	3	0%	33%	100%

Contrastive Translation Pairs

- Goal: find out how well certain translation problems are handled

- Examples
  - noun phrase agreement
  - subject-verb agreement
  - separable verb particle
  - polarity (negative/positive)

- Idea: forced decoding with contrastive translation pair
  - positive example: correct translation
  - negative example: translation with error

- Check if positive example gets better score

Contrastive Translation Pairs



- Noun phrase agreement
  - good: ... *these interesting proposals* ...
  - bad: ... *this interesting proposals* ...
- Subject-verb agreement
  - good: ... *the idea to extend voting rights was* ...
  - bad: ... *the idea to extend voting rights were* ...
- Separable verb prefix
  - good: ... *switch the light on* ...
  - bad: ... *switch the light by* ...

Sennrich (EACL 2017)

- Compares neural machine translation systems for English–German
- Varying word encoding
  - byte pair encoding (BPE)
  - character-based word embeddings (char)
- Results

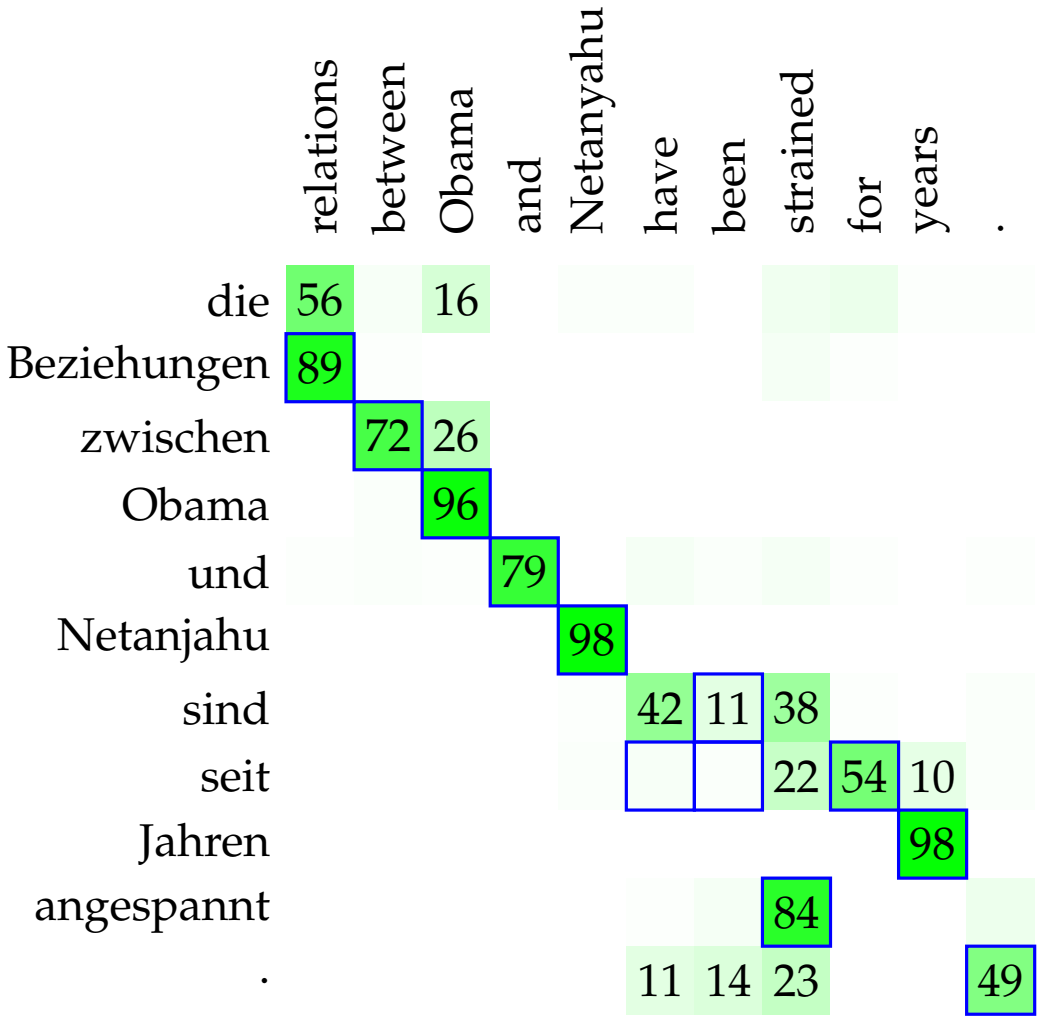
	agreement			polarity (negation)	
system	noun phrase	subject-verb	verb particle	insertion	deletion
BPE-to-BPE	95.6	93.4	91.1	97.9	91.5
BPE-to-char	93.9	91.2	88.0	98.5	88.4
char-to-char	93.9	91.5	86.7	98.5	89.3
human	99.4	99.8	99.8	99.9	98.5

Synthetic Languages

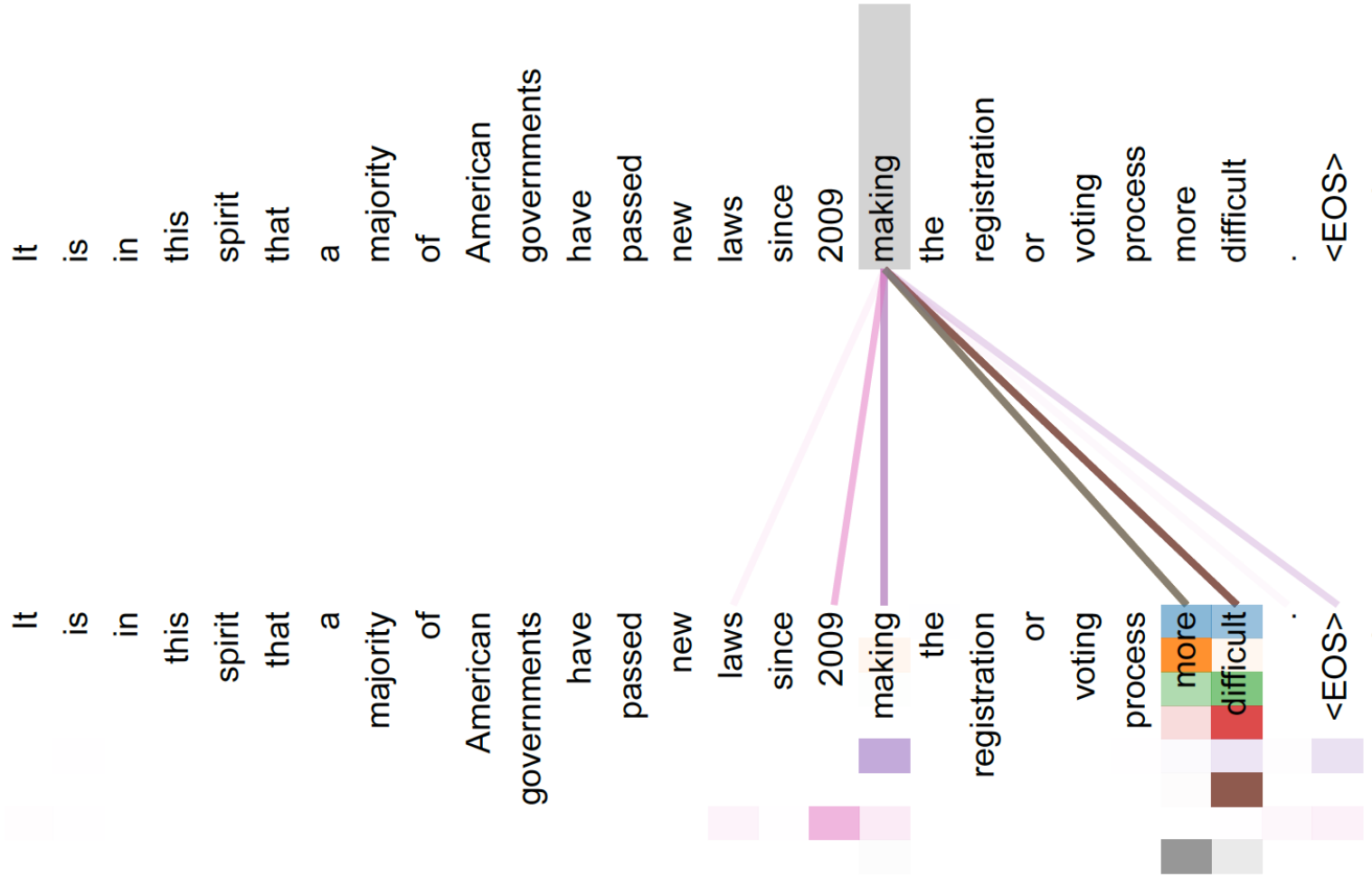
- Create artificial training examples to assess capability of systems
- Example: bracketing language
  - ({ })
  - ({ } { ( ) })
  - { ({ } ({ })) ({ }) }
- Check ability to make correct predictions based on nesting depth, length, etc.

visualization

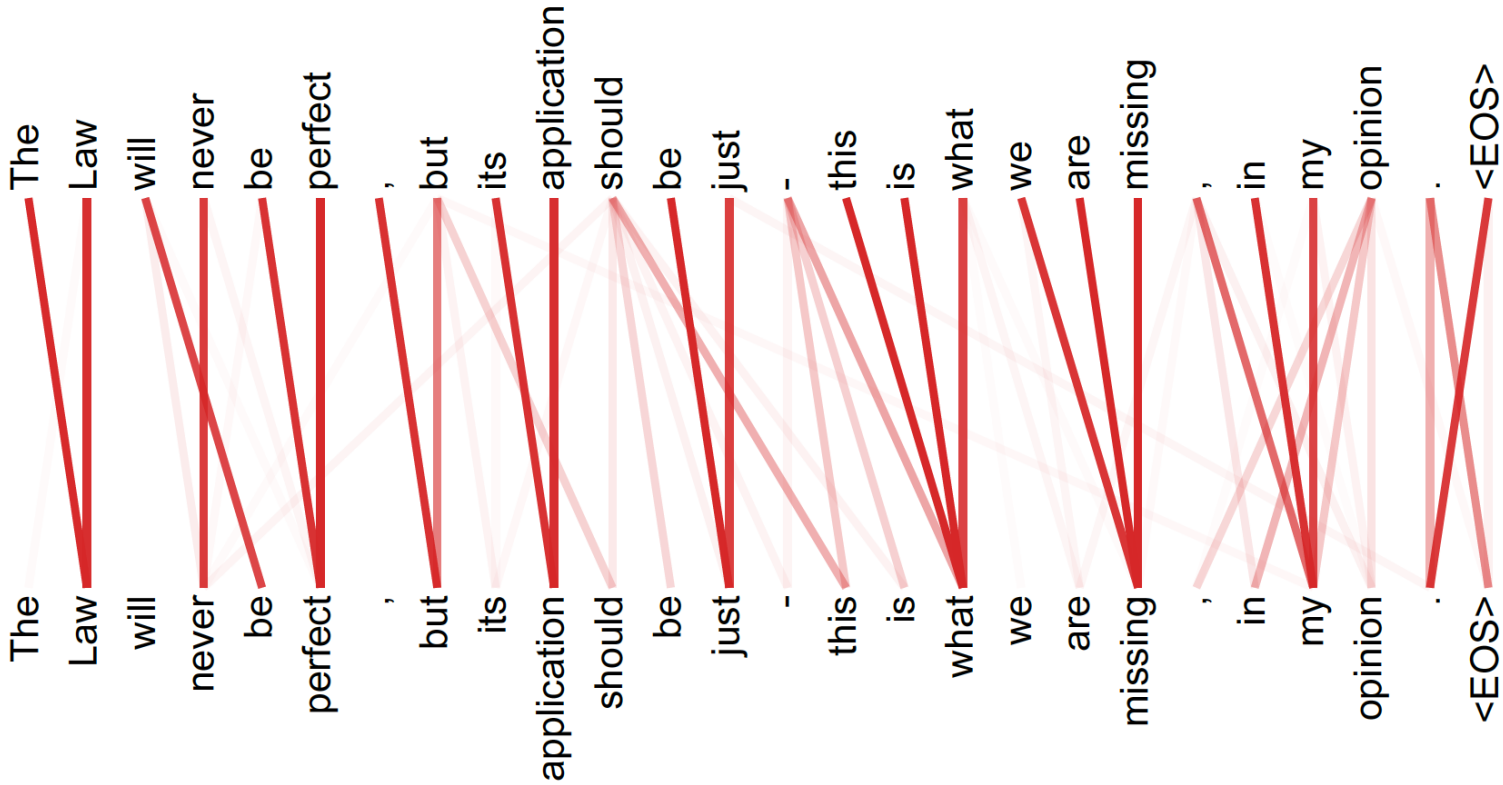
Word Alignment



Multi-Head Attention

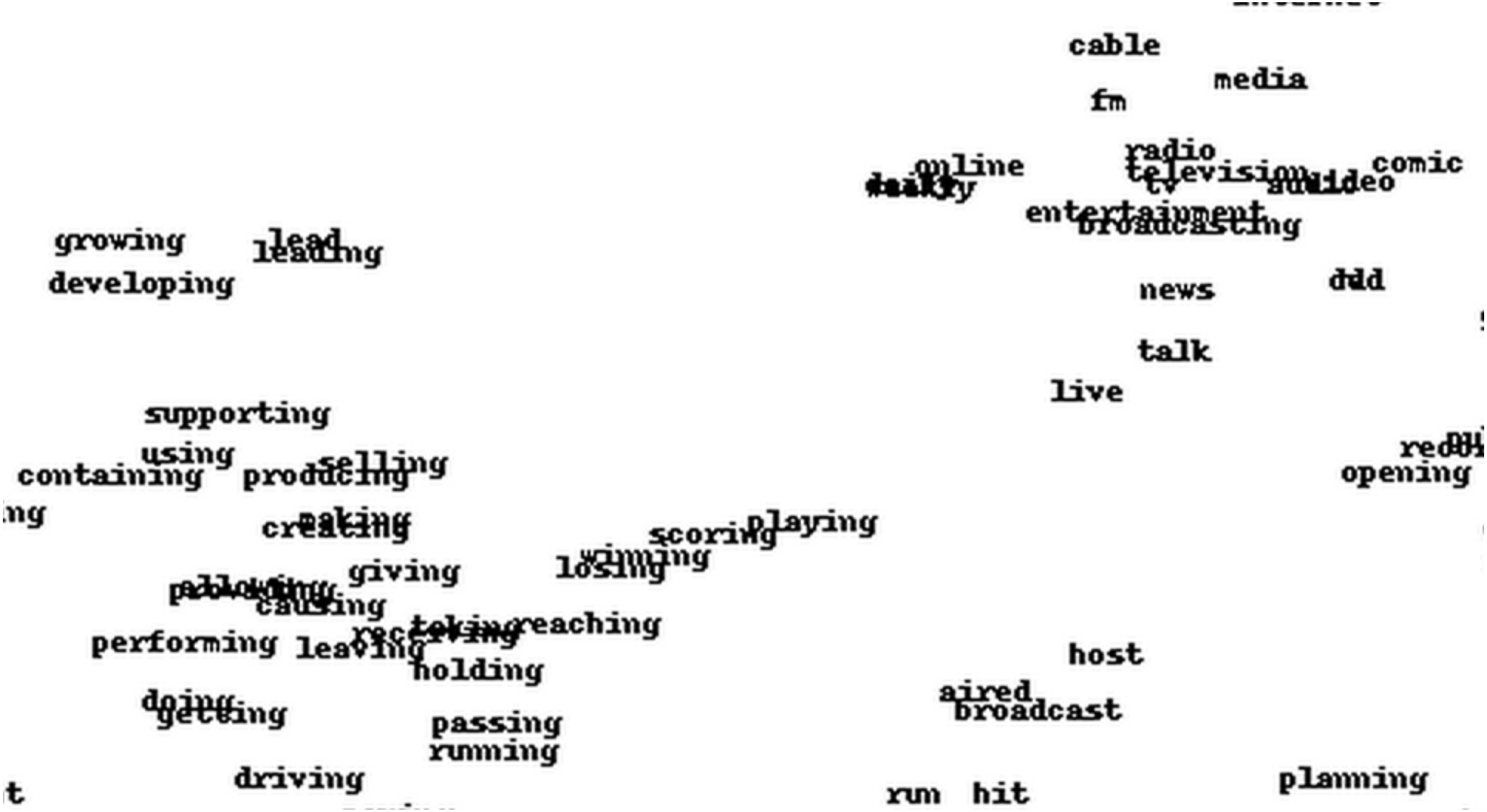


Multi-Head Attention

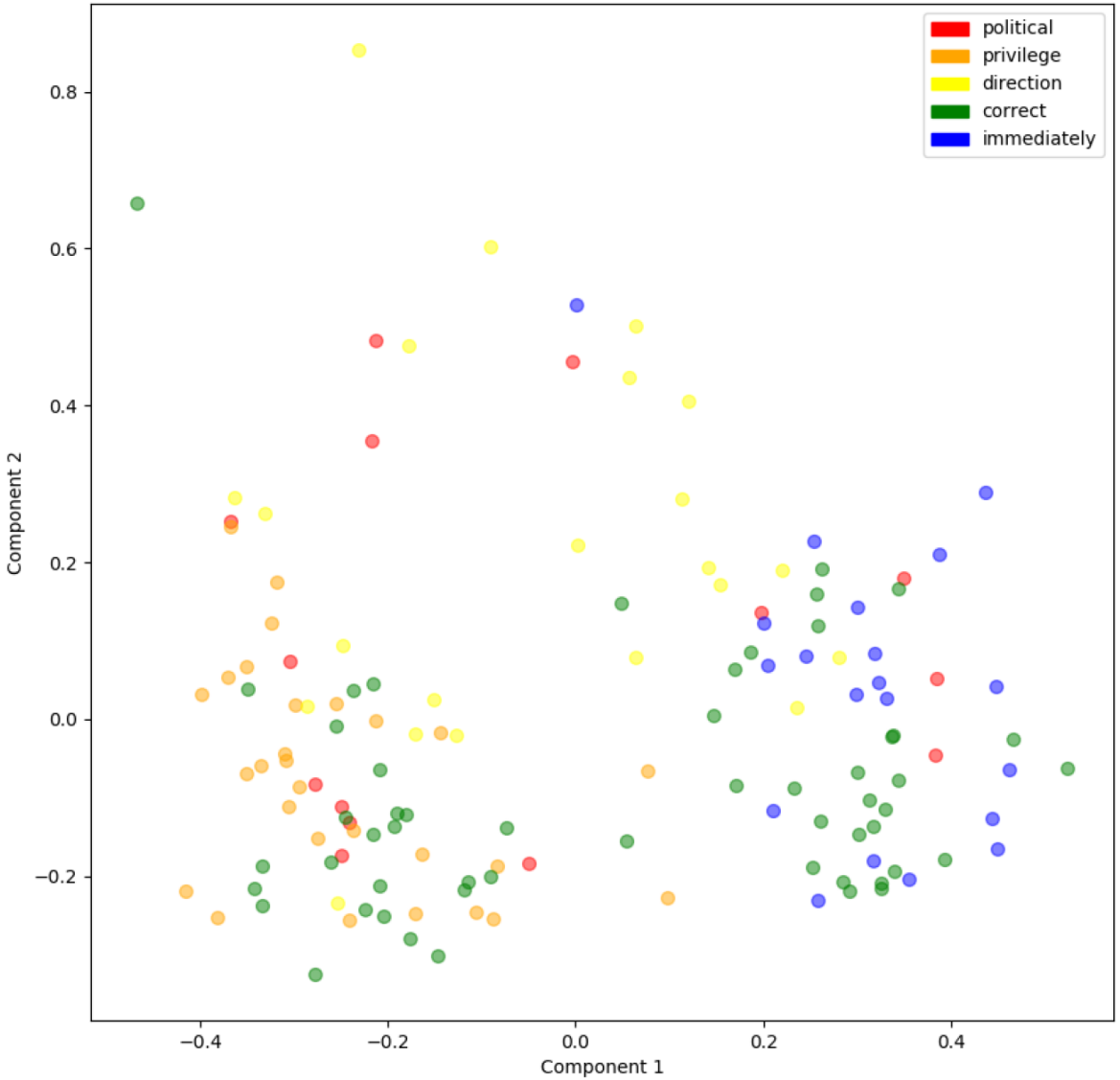


"Many of the attention heads exhibit behaviour that seems related to the structure of the sentence."

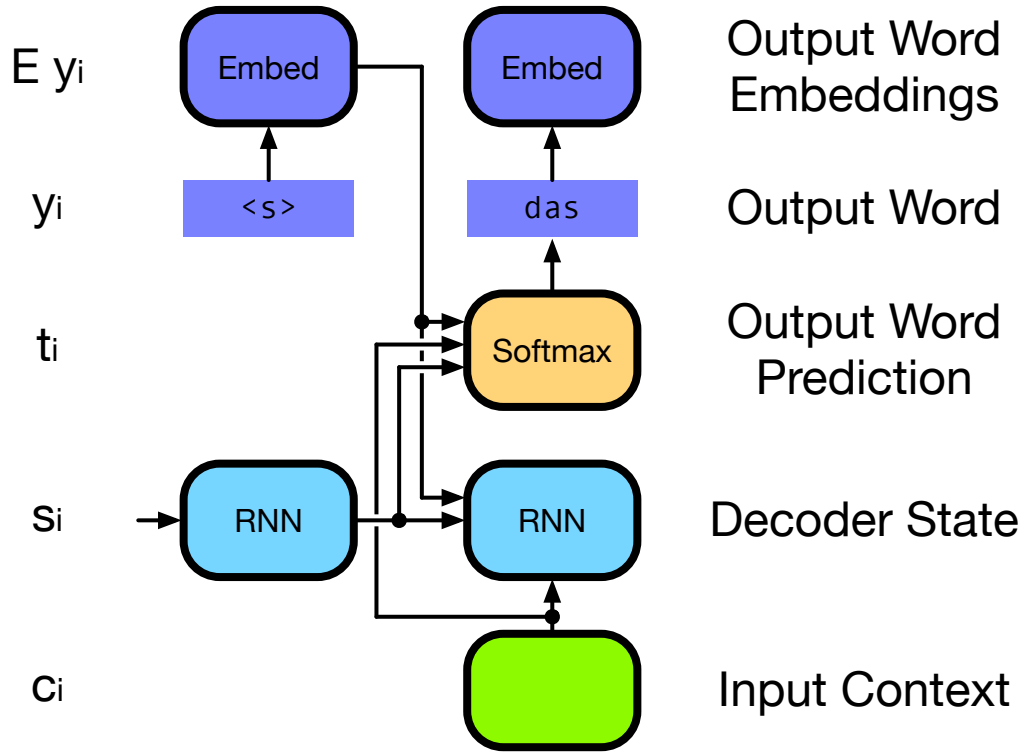
Word Embeddings



Word Sense Clusters



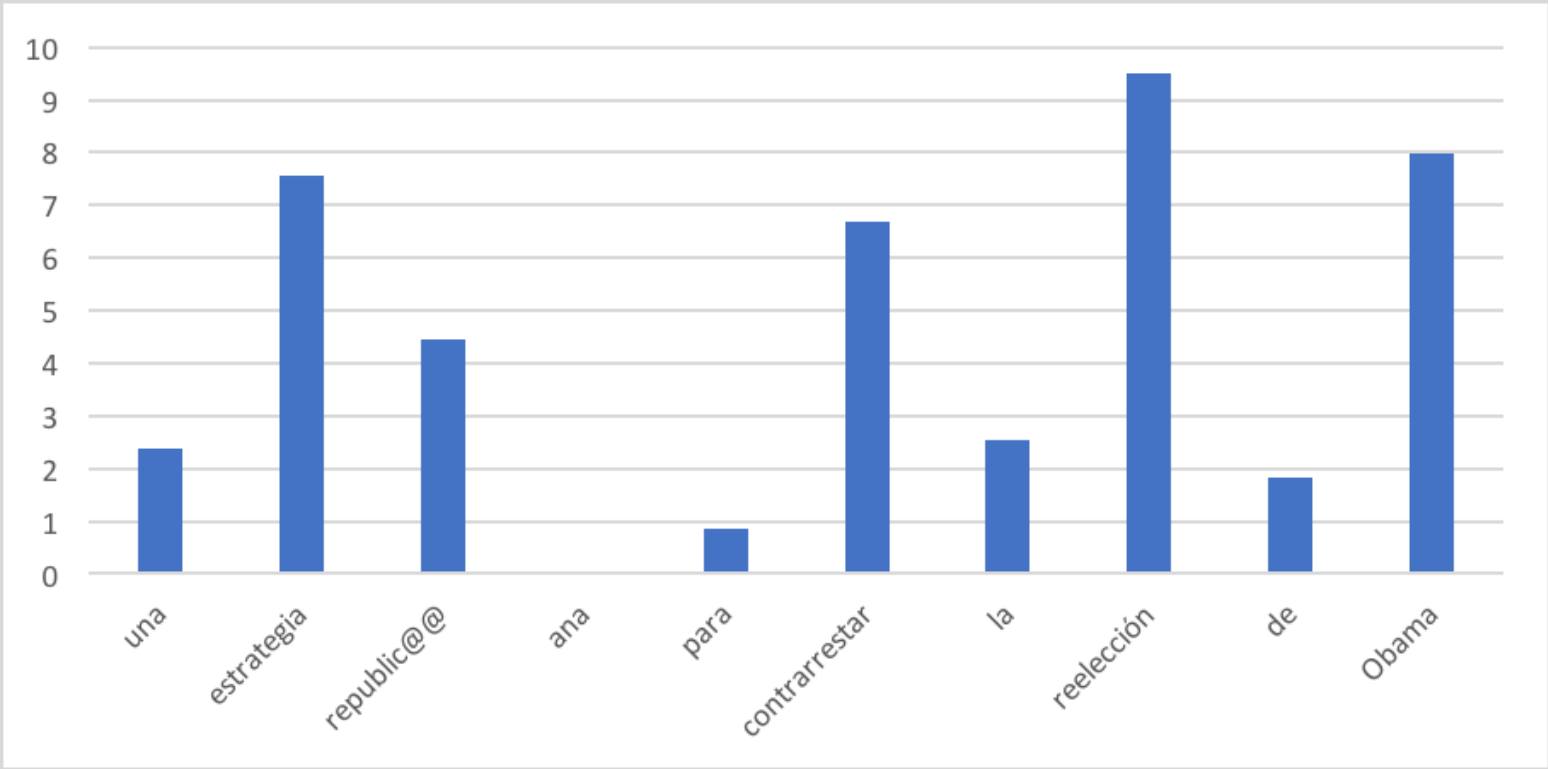
Input Context and Decoder State



- Word predictions are informed by previous output (decoder state) and input
- How much does each contribute?

Input Context vs. Decoder State

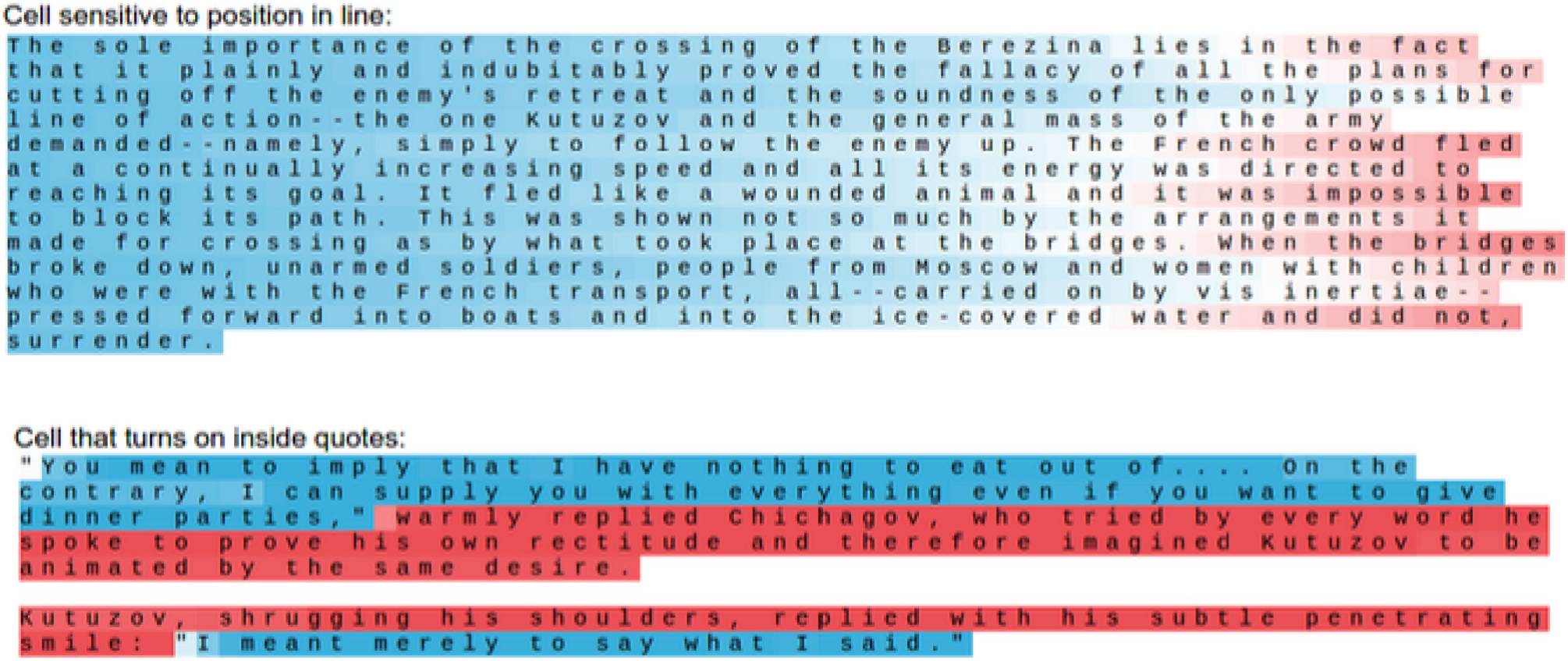
- Input: *Republican strategy to counter the re @-@ election of Obama*



- KL divergence between decoder predictions with and w/o input context
- Input context matters more for content words

identifying neurons

Visualizing Individual Cells



Karpathy et al. (2015): "Visualizing and Understanding Recurrent Networks"

Visualizing Individual Cells

Cell that robustly activates inside if statements:

```
static int __dequeue_signal(struct sigpending *pending, sigset_t *mask,
        siginfo_t *info)
{
  int sig = next_signal(pending, mask);
  if (sig) {
    if (current->notifier) {
      if (sigismember(current->notifier_mask, sig)) {
        if (!(current->notifier)(current->notifier_data)) {
          clear_thread_flag(TIF_SIGPENDING);
          return 0;
        }
      }
    }
    collect_signal(sig, pending, info);
  }
  return sig;
}
```

A large portion of cells are not easily interpretable. Here is a typical example:

```
/* Unpack a filter field's string representation fron user-space
 * buffer. */
char *audit_unpack_string(void **bufp, size_t *remain, size_t len)
{
  char *str;
  if (!*bufp || (len == 0) || (len > *remain))
    return ERR_PTR(-EINVAL);
  /* Of the currently implemented string fields, PATH_MAX
   * defines the longest valid length.
   */
```

Identifying Neurons

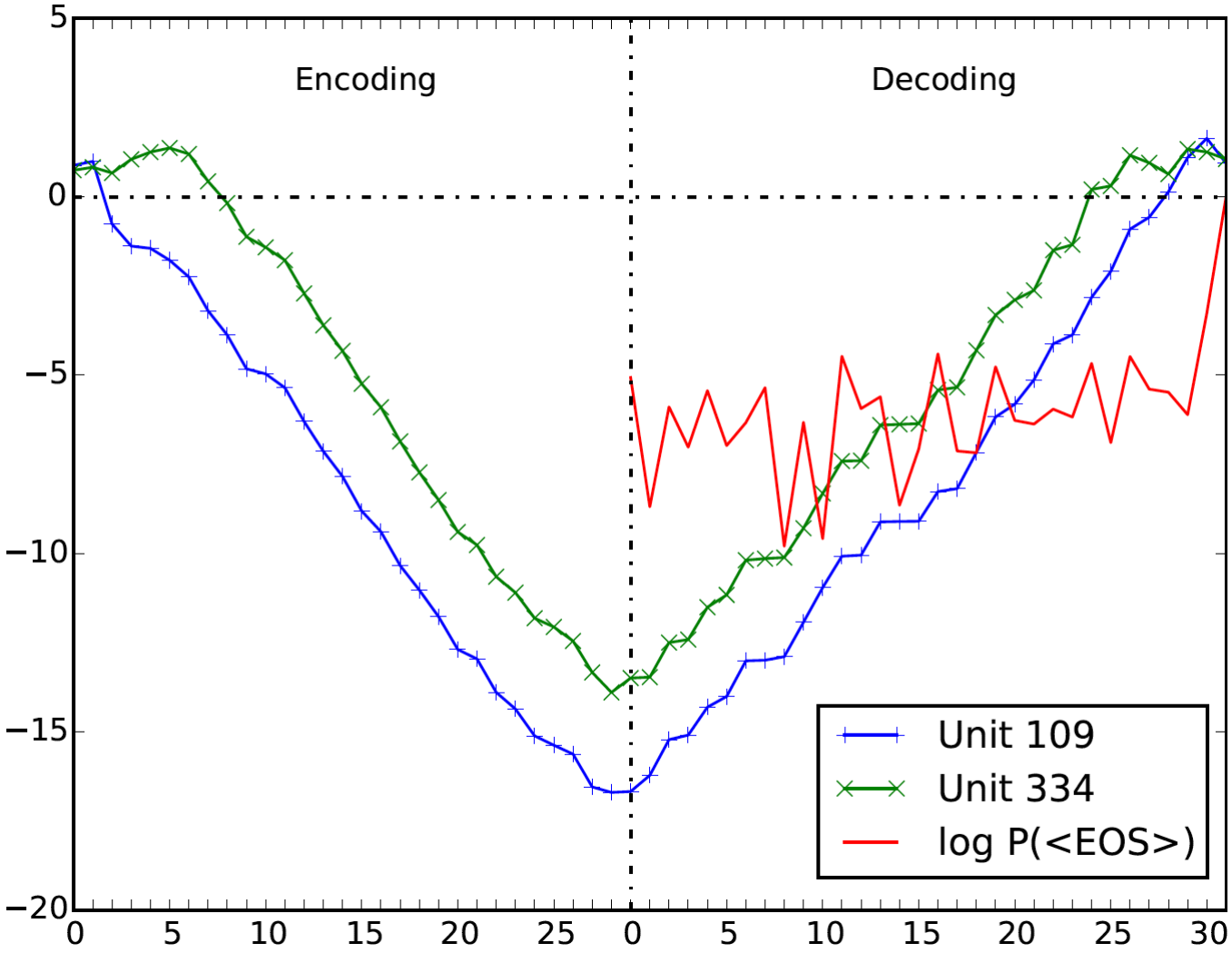


- How are specific properties encoded?
- Easiest case: in a single neuron
- How do we find it?
- Example: length of sequence
  - given: encoder-decoder model without attention
  - does the encoder record the length of the consumed sequence?
  - does the decoder record the length of the generated sequence?

Correlation

- Select a neuron
- Compute correlation
  - value of neuron when processing xth word
  - position x
- Success if highly correlated neuron found

Neurons Correlated with Length



"Why neural translations are the right length" (Shi, Knight, Yuret, EMNLP 2016)

sparse autoencoders

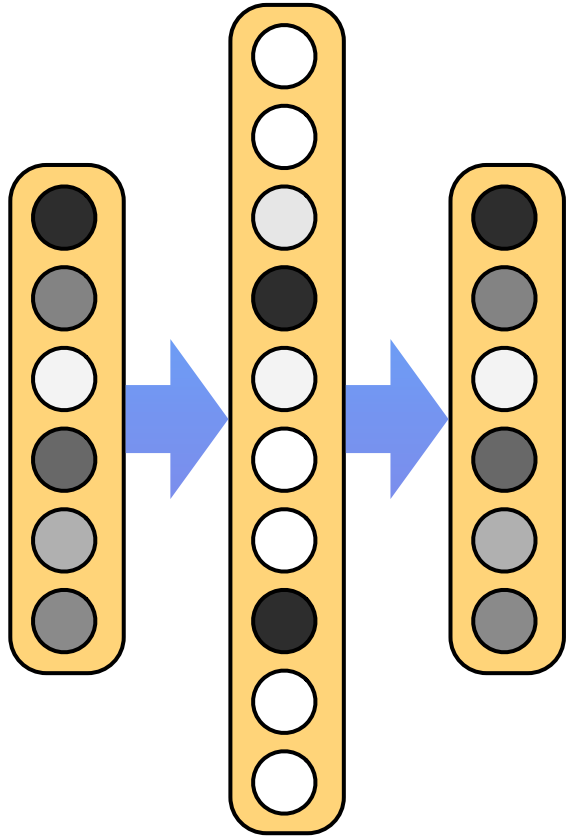
Towards Monosemanticity

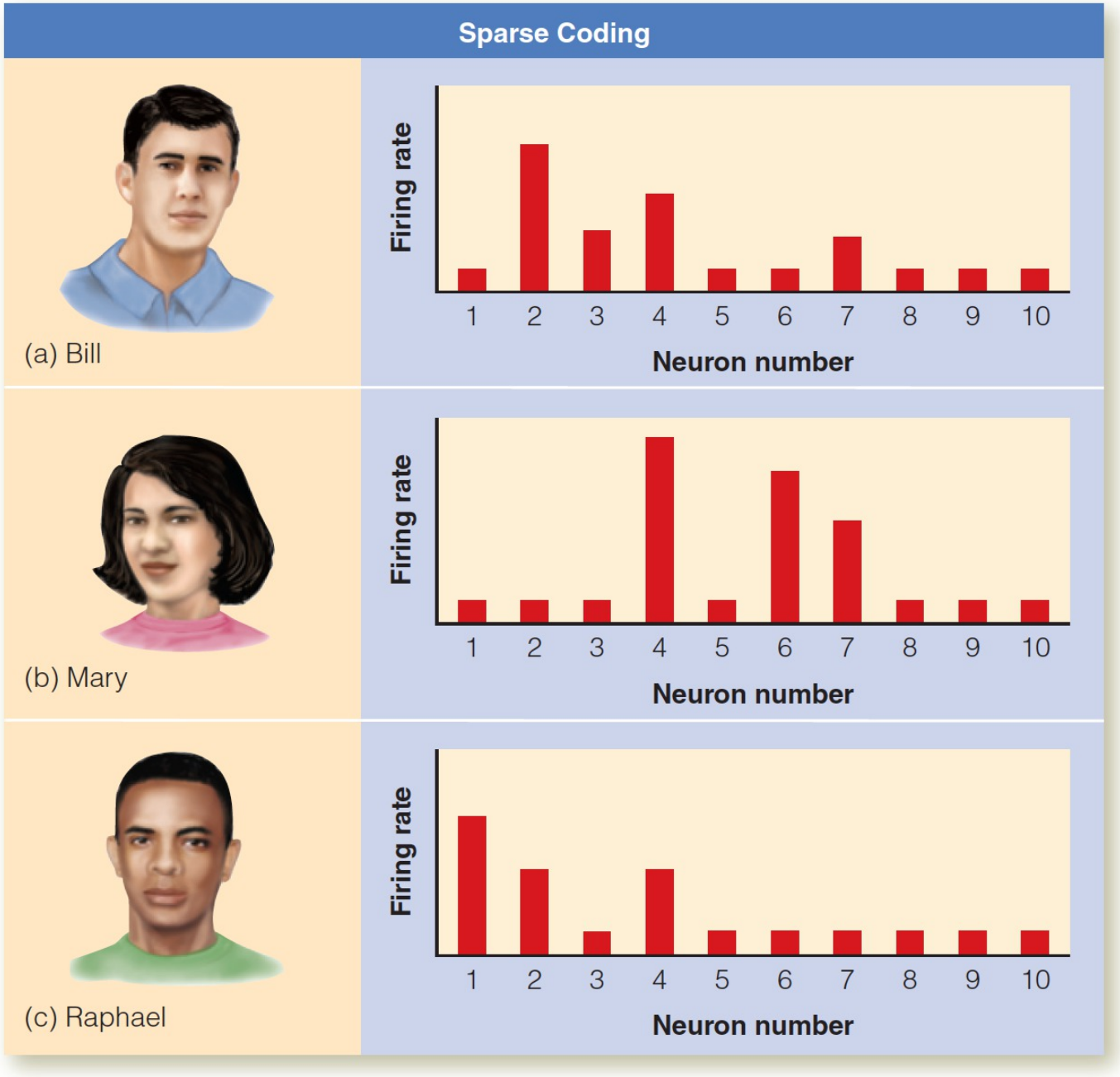
- Individual neurons are **polysemantic**: respond to a mixture of unrelated inputs
- **Superposition**: meaningful features are linear combination of neurons
- How can we detangle them?

$\Rightarrow$ Sparse autoencoders

Sparse Autoencoders

- Take activation for a word in a layer
- Map it into a high dimensional space
  (note: not as usual into a low dimensional space)
- Map back into into original vector
- Two loss term
  - L2 norm of reconstruction loss
  - L1 norm on hidden layer activation




Sparse Coding
(a) Bill
Firing rate
1 2 3 4 5 6 7 8 9 10
Neuron number
(b) Mary
Firing rate
1 2 3 4 5 6 7 8 9 10
Neuron number
(c) Raphael
Firing rate
1 2 3 4 5 6 7 8 9 10
Neuron number

Manual Inspection of Sparse Neurons

Example: Neuron 2281 "DNA (lower case)"



Manual Inspection

- Can be done with different size of hidden layer
  $\rightarrow$ different granularity

- Can be done at different layers

- Cluster sparse hidden layer neurons to get sense of landscape

Online Interface:

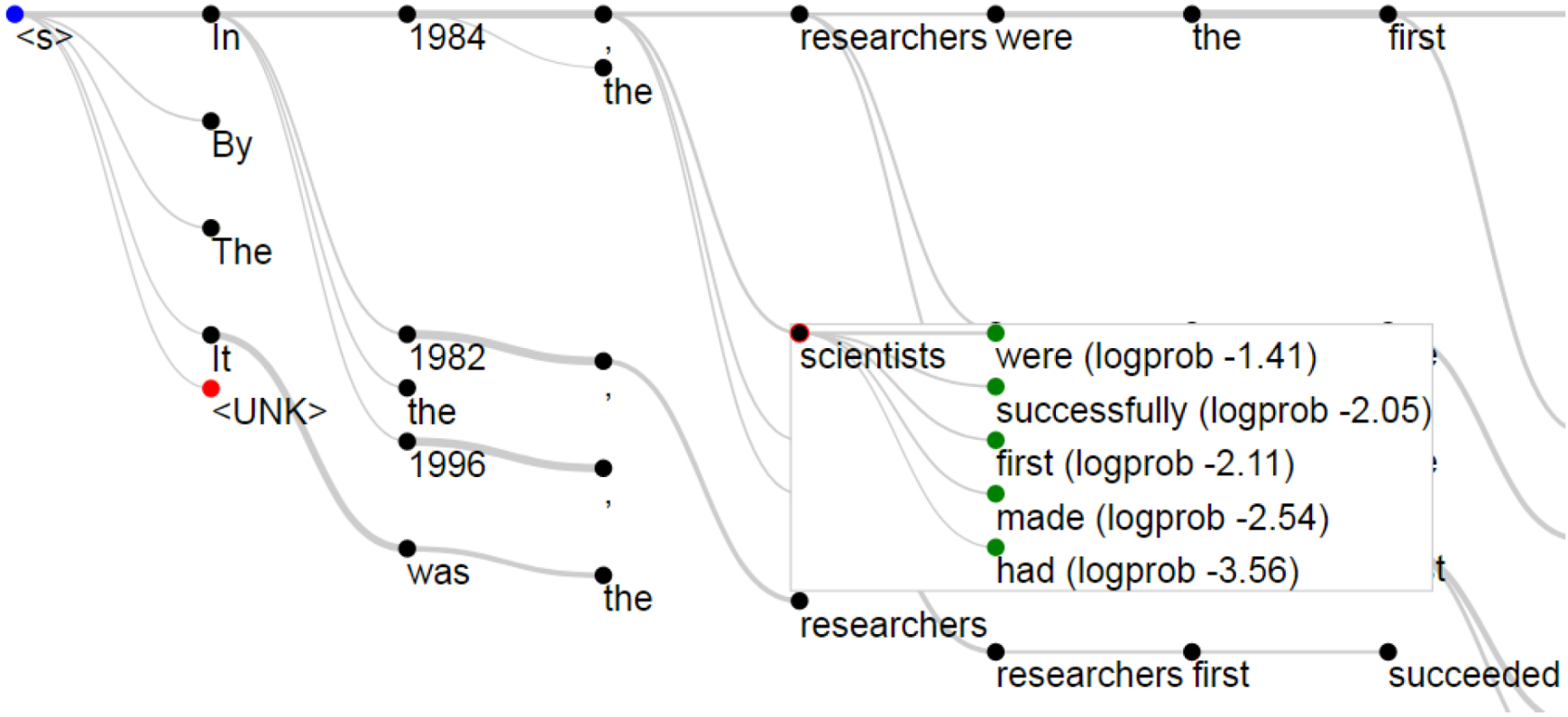
https://transformer-circuits.pub/2023/monosemantic-features/vis/a1.html

visualization tools

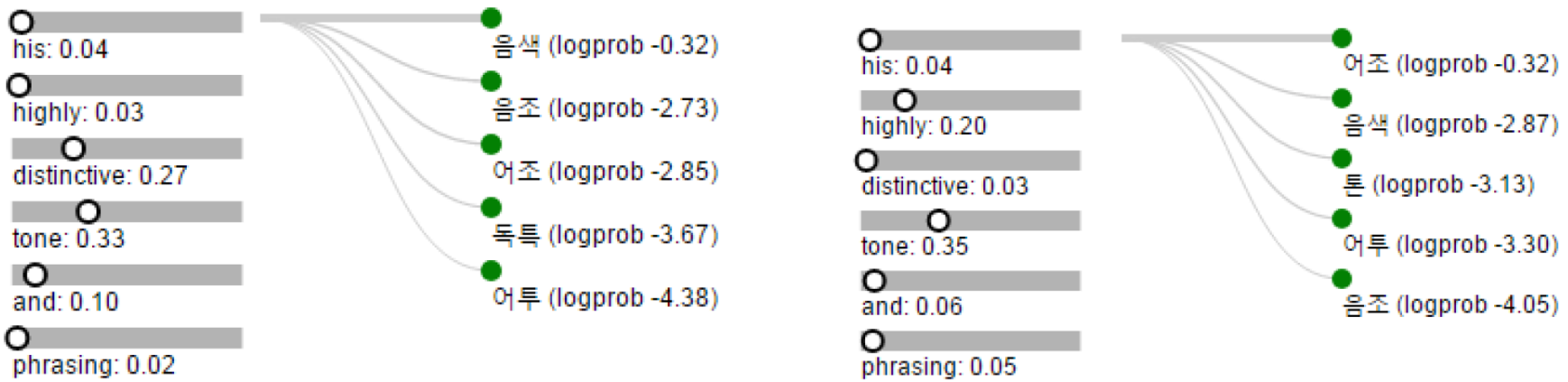
Interactive Exploration

- Tools for inspecting behavior of models and algorithms
- Helps to get insights
- Examples
  - "Interactive Visualization and Manipulation of Attention-based Neural Machine Translation" (Lee et al., EMNLP 2017)
  - "SEQ2SEQ-VIS : A Visual Debugging Tool for Sequence-to-Sequence Models" (Strobelt et al., 2018)

Search Graph

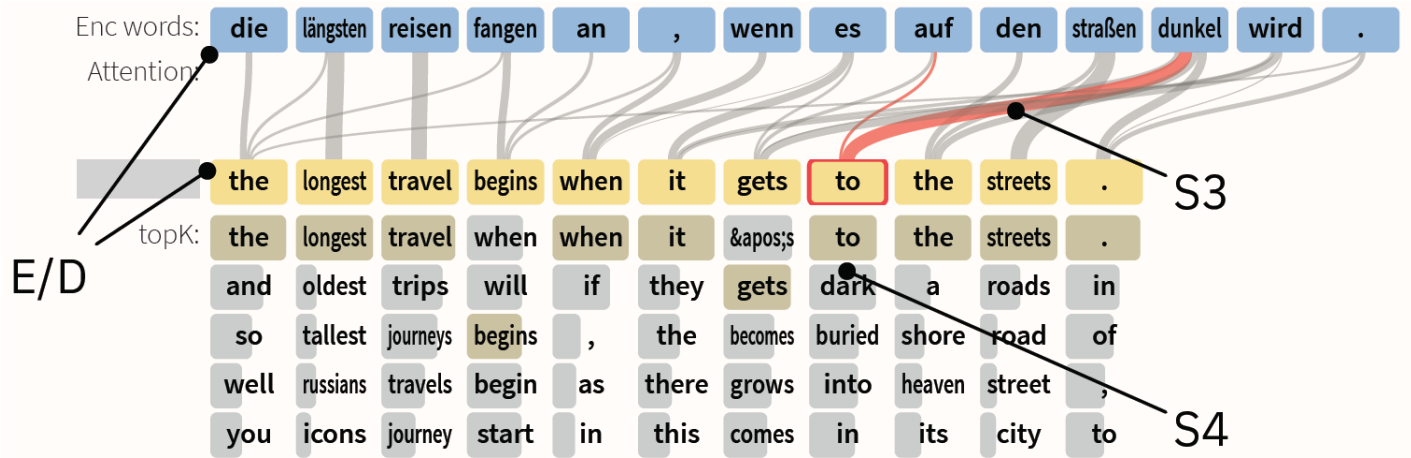


Manipulating Search



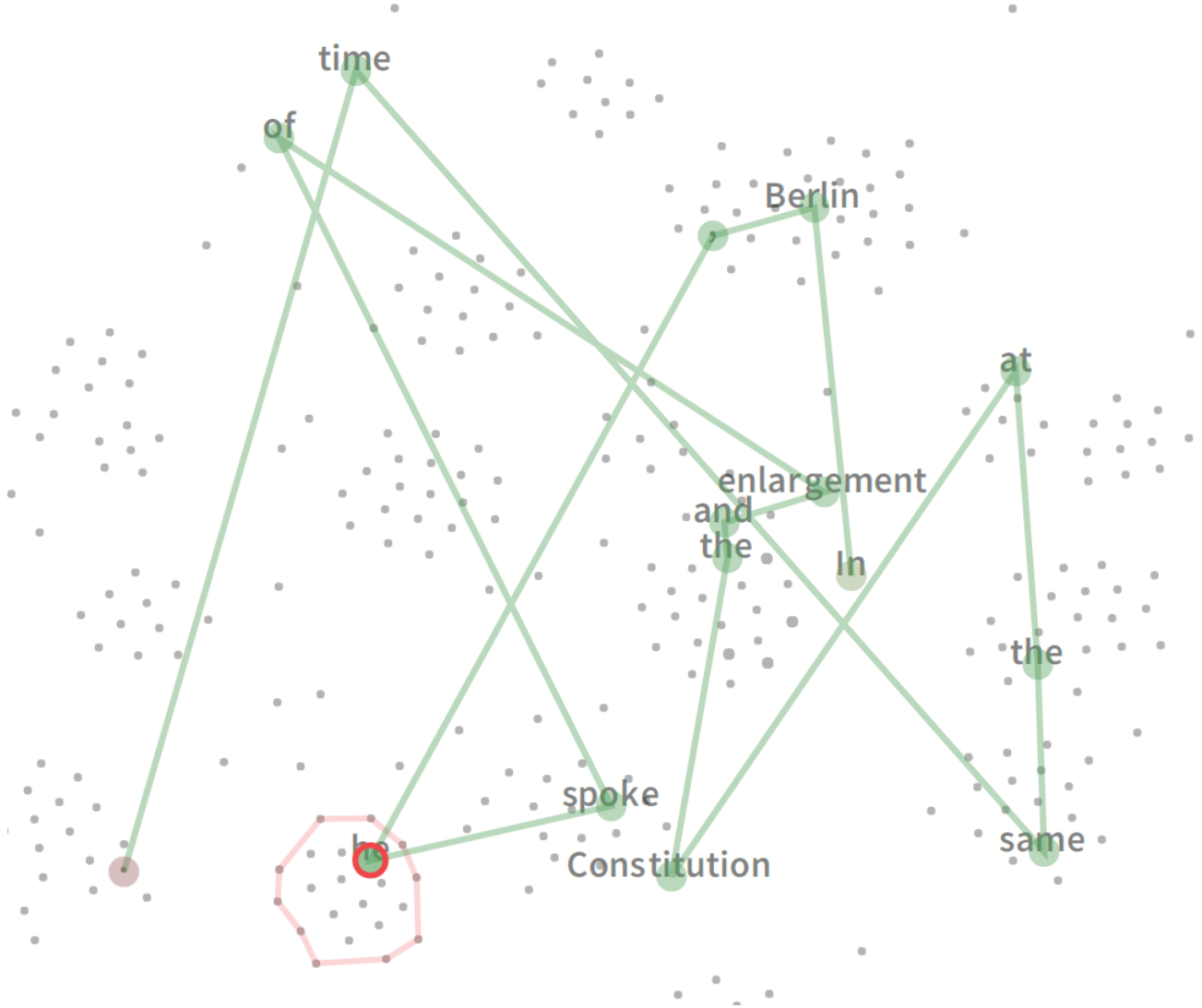
- Inspect attention weights
- Change attention weights → check change in word prediction

Predictions



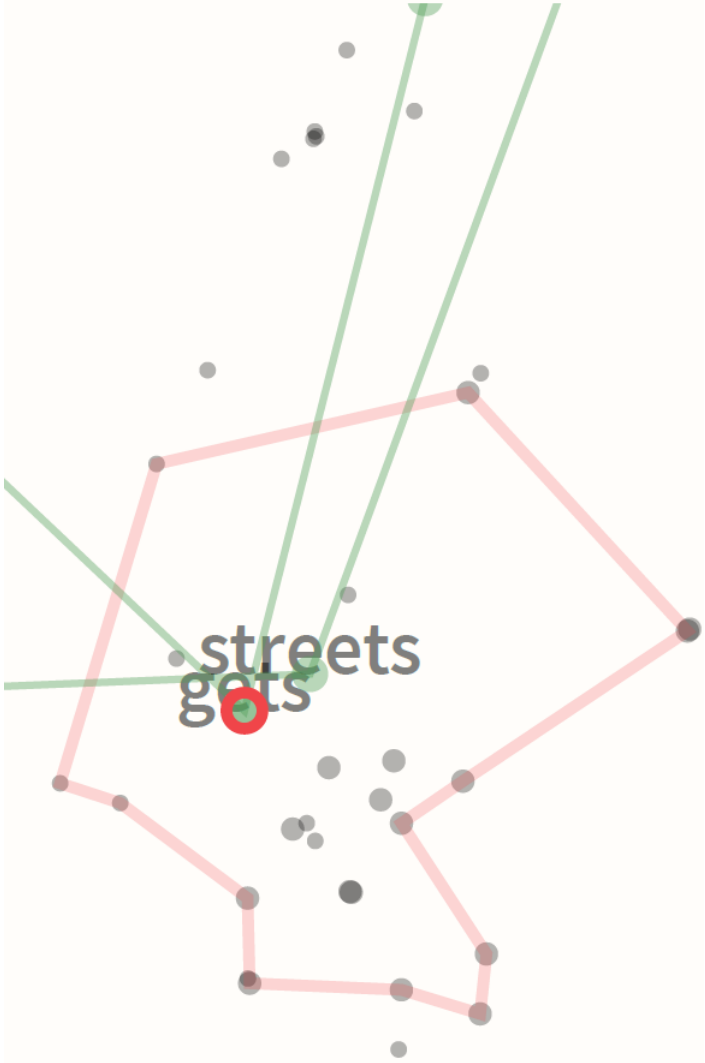
- E/D: encoder and decoder words
- S3: attention weights
- S4: top k predictions

Trajectory of Decoder States

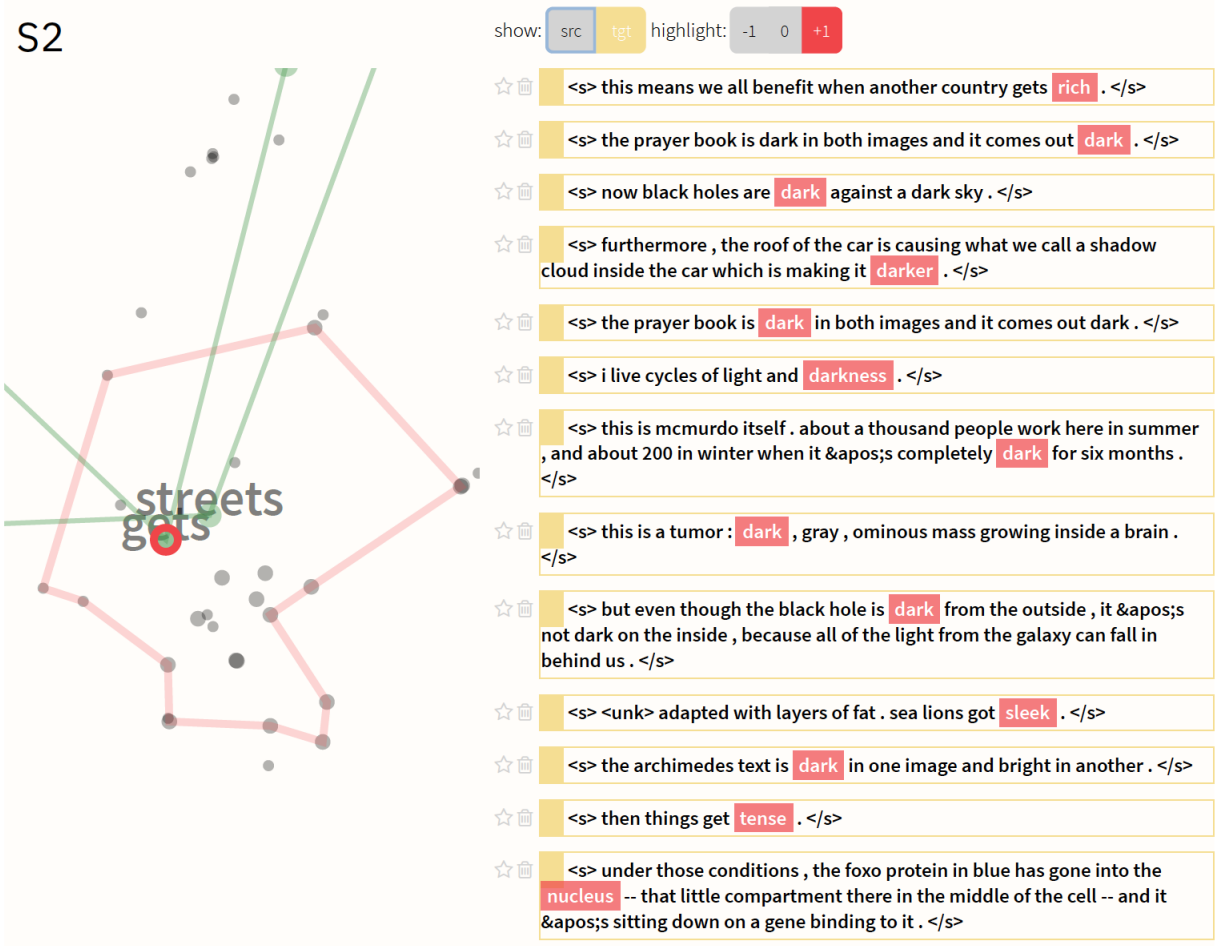


Decoder State Neighborhoods

- 2-D projections of decoder states
- Database of decoder states in training data
- Show neighborhood



Similar Decoder State



logit lens

Lensing Intermediate States

- Transformer models: initial word embeddings processed through several layers
- From the final layer, we predict an output word for each representation
- Can we also inspect intermediate representations?

$\Rightarrow$ Logit Lens: predict output word from each the same way

Lensing Example

- Example: prompt LLM to translate from French to Chinese

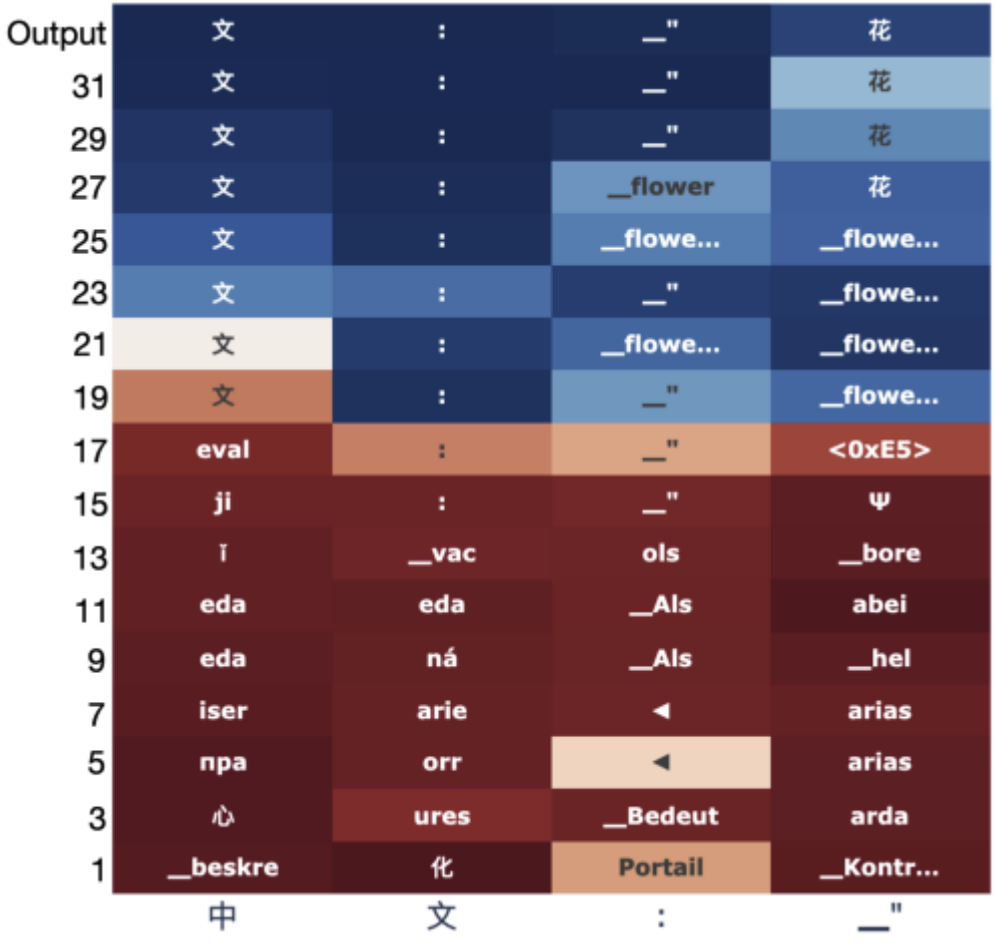
  Fran cois: ”fleur” - 中文: ”

  (中文 means Chinese)

- Inspect intermediate representations

- Claim: Llamas Work in English

  [Wendler et al., 2024]



probing representations

What is in a Representation?



- What is contained in an intermediate representation?
  - word embedding
  - encoder state
  - decoder state

- More specific questions
  - does the model discover morphological properties?
  - does the model disambiguate words?

Classifier Approach

- Pose a hypothesis, e.g.,

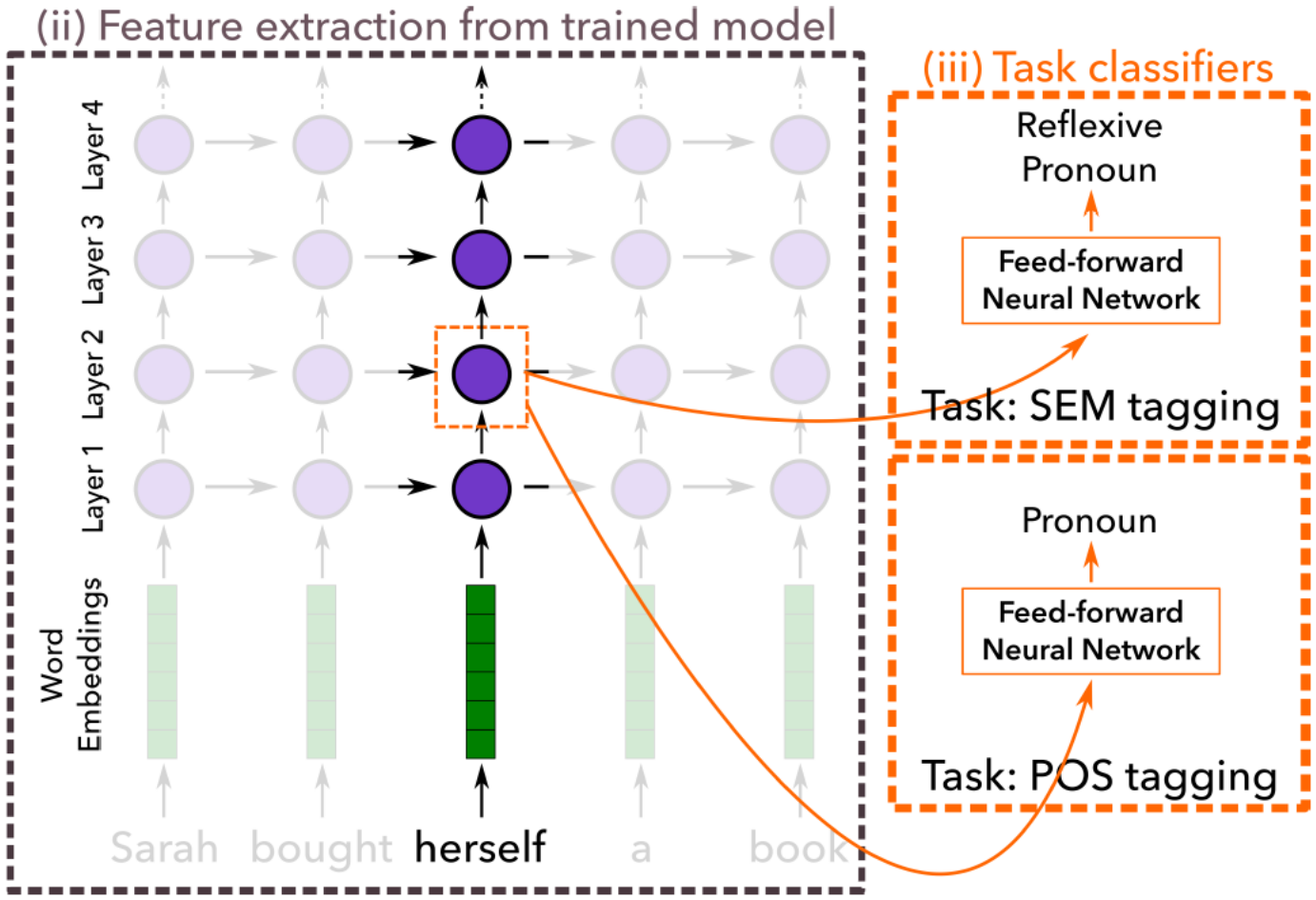
  Encoder states discover part-of-speech.

- Formalize this as a classification problem
  - given: encoder state for word *dog*
  - label: singular noun (NN)

- Train on representations generated by running inference
  - translate sentences not seen during training
  - record their encoder states
  - look up their part of speech tags (running POS tagger or use labeled data)

  → training example (encoder state ; label)

- Test on new sentences



"Evaluating Layers of Representation in Neural Machine Translation on Part-of-Speech and Semantic Tagging Tasks" (Belinkov et al., ACL 2017)

Shi et al. (EMNLP 2016)

- LSTM sequence-to-sequence model without attention

- Different tasks
  - translate English into Russian, German
  - copy English
  - copy permuted English
  - parse English into linearized parse structure

- Predict
  - constituent phrase (NP, VP, etc.)
  - passive voice and tense

- Findings
  - much better quality when translating than majority class
  - same quality for copying as majority class

Belinkov et al. (EMNLP 2017)

- Attentional neural machine translation model
- Predict
  - part-of-speech tag
  - semantic tag
    - type of named entity
    - semantic relationships
    - discourse relationships
- Findings
  - compare prediction quality of different encoder layers
  - mostly better performance at deeper layers
  - little impact from target language

Belinkov et al. (ACL 2017)



- Attentional neural machine translation model with character-based word embeddings

- Predict for morphologically rich input languages
  - part-of-speech tag
  - morphological properties

- Findings
  - character-based representations much better for learning morphology
  - word-based models are sufficient for learning structure of common words
  - lower layers better at word structure, deeper layers better at word meaning
  - target language matters for what information is learned
  - neural decoder learns very little about word structure

relevance propagation

What Determined Output Decision?



- What part of the network had the biggest impact on final decision?

- For instance machine translation:
  - prediction of a specific output word
  - which of the input words mattered most?
  - which of the previous output words mattered most?

Layer-Wise Relevance Propagation

- Start with output prediction

  i.e., high value for word in softmax

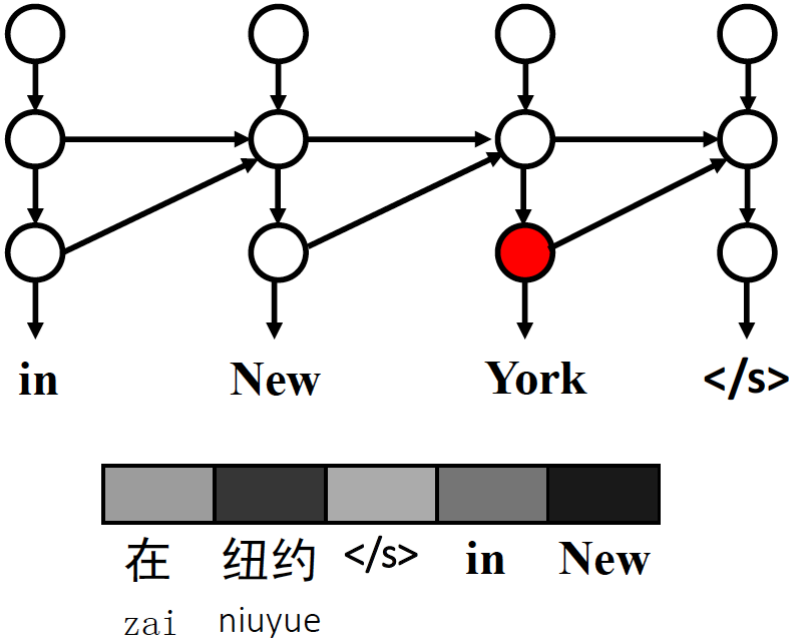
- Compute backwards what contributed to this high value

- First step

  - consider values of previous layer
  - consider weights from previous layer
  - assign relevance values for each node in previous layer
  - normalize so they add up to one

- Recurse until input layer is reached

Example: Chinese–English



"Visualizing and Understanding Neural Machine Translation"
(Ding, Liu, Luang and Sun, ACL 2017)

saliency

Saliency

- Intuition

if a decision changes a lot if a specific input value changes

⇓

more relevant

change in the input value has no impact on decision

⇓

not relevant

- Mathematically
  - relationship $p(y_0|x_0)$ between an input value x_0 and an output value y_0
  - assume this to be a linear relationship (which is approximately true locally)
  - compute slope by derivative

$$\text{saliency}(x, y) = \frac{\partial p(y|x)}{\partial x}$$

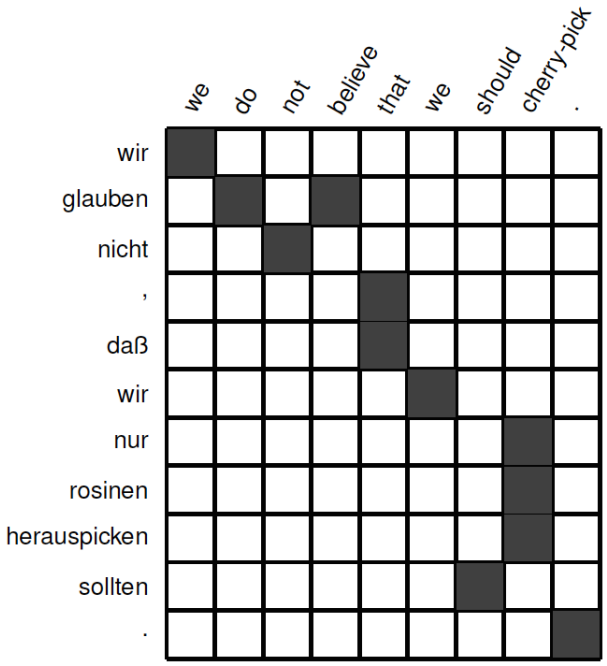
Example: Word Alignment

- Which input word had the most influence on an output word prediction?

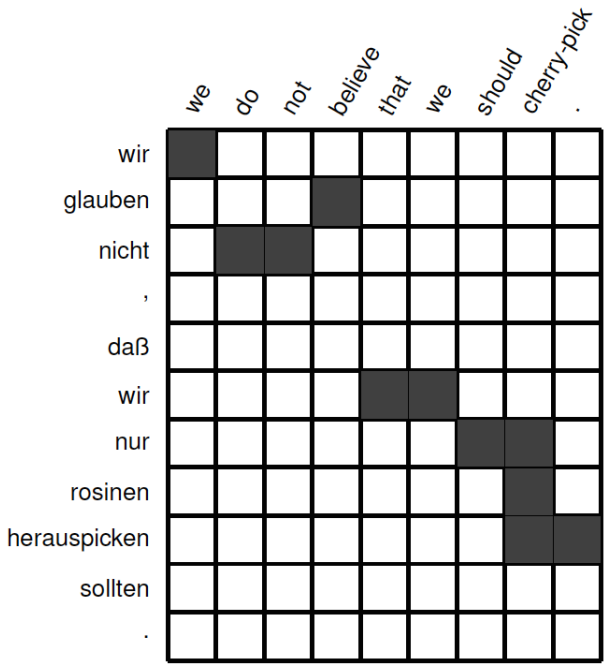
$\Rightarrow$ Trace back to word embeddings

- Note
  - not interested in individual neurons
  - combine salience values in embedding vector

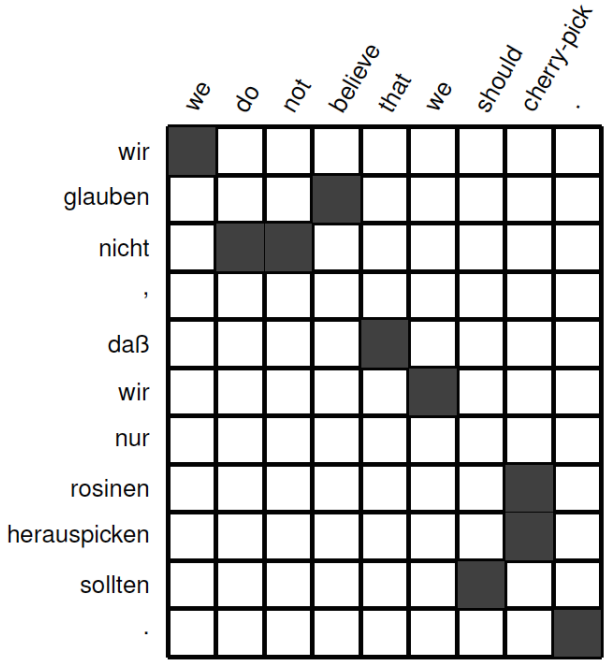
Saliency



Human Reference



Attention



Saliency

What Do Interpretability Measures Reveal?



- How are do we know if these methods are doing the right thing

what a model should be doing
$\neq$
what a model is doing

- Also: impact of input word $\neq$ word alignment
  - *bank* most responsible to produce German translation *Bank*
  - *credit* or *account* may be crucial for word sense disambiguation
  - other words may provide clues that word is a noun (not a verb)

Explainable AI

- Important question for users

Why did the network reach this decision?

- Tracing back decisions to inputs

$\Rightarrow$ Causal explanation



questions?